MIL-B-3493D
30 September 1985
SUPERSEDING
MIL-B-3493C
29 May 1967

MILITARY SPECIFICATION

BUCKLES, END BAR WITH TONGUE (FOR LEATHER)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers one type of end bar buckle with roller and tongue.

* 1.2 Classification. Buckles shall be of the following classes and sizes as shown on Drawing 4-1-30 (see 6.2):

Class 1 - Copper alloy (brass)
Class 3 - Steel wire

2. APPLICABLE DOCUMENTS

* 2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data that may be of use in improving this document should be addressed to U.S. Army Natick Research and Development Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5340

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SPECIFICATIONS

FEDERAL

QQ-B-613	- Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
QQ-C-390	- Copper Alloy Castings (Including Cast Bar)
QQ-N-290	- Nickel Plating (Electrodeposited)
QQ-S-698	- Steel, Sheet and Strip, Low Carbon
QQ-W-321	- Wire, Copper Alloy
QQ-W-461	- Wire, Steel, Carbon (Round, Bare, and Coated)
TT-E-529	- Enamel, Alkyd, Semigloss
PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-676	- Boxes, Setup

MILITARY

MIL-F-495	- Finish, Chemical, Black for Copper-Alloys
MIL-L-10547	- Liners, Case, and Sheet, Overwrap; Water - Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Products
MIL-STD-147	- Palletized Unit Loads

DRAWING

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

4-1-30	- Buckles, End Bar, with Tongue (For Leather)

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel
D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

* 3.1 Materials. The materials specified shall conform to applicable document and requirements specified herein. Recycled material may be used (see 6.3).

3.1.1 Brass.

* 3.1.1.1 Copper alloy (brass) castings. Copper alloy (brass) castings shall conform to alloy number A3 of QQ-C-390.

* 3.1.1.2 Brass sheet or strip. Brass sheet or strip shall conform to alloy numbers 260 or 268, quarter-hard temper, of QQ-B-613.

* 3.1.1.3 Copper alloy wire. Copper alloy wire shall conform to alloy number 260 or 270, quarter-hard temper, of QQ-W-321.

3.1.2 Steel.

3.1.2.1 Steel wire. Steel wire shall conform to steel number 1006-1010, finish 1, annealed in process of QQ-W-461.

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* 3.1.2.2 Steel, sheet and strip. Steel sheet shall conform to hot rolled or cold rolled, commercial quality; steel strip shall conform to cold rolled any temper of QQ-S-698.

* 3.1.3 Enamel. Enamel shall be black conforming to type II of TT-E-529.

3.2 Design and construction. The design, construction, and dimensions of the buckles shall conform to Drawing 4-1-30 and the requirements specified herein.

3.2.1 Buckle body.

3.2.1.1 Class 1. Body of class 1 buckle shall be cast from brass specified in 3.1.1.1.

3.2.1.2 Wire, body, class 3. Body of class 3 buckle shall be formed from one piece of wire specified in 3.1.2.1 and welded. Welds shall be sound, smooth, not porous and free from holes, flash, cracks, burned, or reduced area when examined as specified in 4.2.3.1. Welds shall be located as shown on Drawing 4-1-30.

* 3.2.2 Tongues for buckles class 1 and class 3. Tongues for class 1 buckles shall be fabricated from copper alloy wire specified in 3.1.1.3. Tongues for class 3 buckles shall be fabricated from steel wire specified in 3.1.2.1. After assembling, the tongue shall not bind on the buckle body heel or slip out of heel groove or channel and the tip shall not protrude beyond front of roller but lie in close contact with roller.

3.2.3 Rollers. Rollers for buckles, after assembly, shall rotate freely without binding on buckle body. The joint of the roller along its entire length shall be butted evenly.

3.2.3.1 Class 1. Rollers for class 1 buckle shall be fabricated from quarter-hard temper brass sheet or strip specified in 3.1.1.2.

3.2.3.2 Class 3. Rollers for class 3 buckle shall be fabricated from steel sheet or strip specified in 3.1.2.2.

3.3 Finish. Prior to application of finish, all buckles shall be free from corrosion, rust, dirt, and discoloration.

3.3.1 Buckles, class 1. Class 1 buckles shall be given a black chemical finish conforming to MIL-F-495 and tested as specified in 4.2.4.3.

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* 3.3.2 Buckles, class 3. When specified (see 6.2), class 3 buckles shall be nickel plated in accordance with class 1, type II of QQ-N-290.

3.4 Marking for identification. The buckles shall be permanently marked with the letters "US" and the manufacturer's identification, where possible, in accordance with MIL-STD-130. The marking shall be made on the back surface of the buckle body in such a manner that the front surface shall show no signs of penetration, uneveness, or cause distortion on any part of the buckle.

3.5 Workmanship. The buckles shall be free from digs or gouges, burrs, slivers, sharp edges, dents, tool marks, corrosion, rust, dirt, discoloration, or areas of no finish.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents.

4.2.2 Process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether the required operations are accomplished as specified. When applicable, process inspection shall be conducted to see that class 3 buckles have zinc-plate applied before enameling.

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4.2.3 Intermediate inspection.

4.2.3.1 Intermediate visual examination of welded class 3 buckles. Class 3 welded buckles shall be examined before finishing and assembling the roller to determine conformance with 3.2.1.2. The lot shall be all unfinished buckles of one size offered for inspection at one time prior to assembling the roller. The sample unit for this examination shall be one buckle. The inspection level shall be S-1 with an acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units.

4.2.4 End item inspection. The buckles shall be examined in accordance with 4.2.4.1 and 4.2.4.2. The lot shall consist of all completely finished buckles of one class and size offered for inspection at one time. The sample unit for this inspection shall be one completely fabricated and finished buckle.

4.2.4.1 Visual inspection. The buckles shall be examined in accordance with the classification of defects in table I. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 total defects expressed in terms of defects per hundred units.

TABLE I. Classification of defects

Examine	Defect	Category	
		Major	Minor
Finish	Not type specified	101	
General	Evidence of corrosion	102	
Black chemical (when applicable)	Color not as specified		201
	Finish not clean and uniform		202
	Area of no finish or finish discolored		203

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TABLE I. Classification of defects - Continued

Examine	Defect	Category	
		Major	Minor
Enamel (when applicable)	Area of no film or thin film or color not as specified	103	
	Separation of color or finish rough (i.e., lint, dust, dirt or other foreign matter imbedded in finish)		204
	Finish not continuous, smooth and adherent (i.e., orange peel, wrinkles, drops, streaks)		205
Nickel plated (when applicable)	Coating omitted or area of no coating	104	
	Coating not smooth, uniform, fine-grained or not tightly adherent (i.e., flaky, blistered, shaded, broken, or cut through, or is not free from pits, nodules, or indications of burning		206
	Not bright finish	105	
Construction and workmanship (all classes and sizes as applicable)	Wrong size or class	106	
	Sharp edge or projections that may cause injury	107	
	Cracked, broken, bent, malformed, or damaged	108	
	Pronounced tool or forming marks		207
	Castings miscast, incomplete, or contains blowholes or evidence of porosity	109	
	Roller does not turn freely on body	110	
	Roller bent, but turns freely		208
	Roller joint not butted	111	
	Tongue mounted in reverse position	112	
	Tongue binds on body heel		209
	Tongue will not move on body heel	113	
	Tip of tongue protrudes beyond edge of roller		210

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TABLE I. Classification of defects - Continued

Examine	Defect	Category	
		Major	Minor
Construction and workmanship (all classes and sizes as applicable (Continued)	Tip of tongue not in close contact with body		211
	Tongue moves laterally on body (i.e., slips out of groove or channel)	114	
	Any component part missing	115	
Identification marking	Missing, incomplete, illegible, or incorrect		212
	Not on back surface of buckle		213
	Indication of penetration, unevenness, or distortion on face surface of buckle		214

4.2.4.2 Dimensional inspection. Inspection shall be made of buckles for compliance with dimensions specified. Any dimension that is not within the specified tolerances shall be classifed as a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

4.2.4.3 End item testing. Testing of the completely finished class 1 buckle shall be performed in accordance with table II for the characteristics shown therein. The inspection level shall be S-1 and the AQL shall be 6.5, expressed in terms of defects per hundred units.

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TABLE II. End item testing

Characteristic	Requirement paragraph	Test Method	Number determinations per sample unit	Results reported as
Black chemical finish, class 1 buckles	3.3.1			
Color and gloss		MIL-F-495	1	Pass or fail
Resistance to hot soap solution		MIL-F-495	1	Pass or fail
Resistance to accelerated weathering		MIL-F-495	1	Pass or fail

4.2.5 Examination for count in intermediate containers. Intermediate containers shall be examined to determine conformance with unit pack markings and specified quantity. The lot shall be the number of intermediate containers submitted for inspection. The sample unit for this examination shall be one container (unit pack). Any container with less than the specified or marked quantity of buckles shall be classified as a defect. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

* 4.2.6 Packaging inspection. An examination shall be made to determine compliance with preservation, packing, and marking requirements of section 5. Defects shall be scored in accordance with table III. The lot shall be the number of shipping containers offered for inspection. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

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TABLE III. Packaging inspection

Examine	Defect
Marking (exterior and interior container)	Omitted; incorrect, illegible; of improper size, location, sequence, or method of application
Material	Component missing, damaged, or otherwise defective
Contents (exterior)	Number per container is more or less than required

* 4.2.7 Palletization inspection. An inspection shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list in table IV. The lot size shall be the number of palletized unit loads in the end item inspection lot. The sample unit shall be one palletized unit load fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE IV. Palletization inspection

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified

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TABLE IV. Palletization inspection - Continued

Examine	Defect
Weight	Exceeds maximum load limits
Marking	Omitted, incorrect, illegible, or of improper size, location, sequence, or method of application

* 5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), buckles of one class and size only shall be unit packed in quantities specified in table V in a snug-fitting fiberboard box conforming to style RSC, type CF (variety SW) or type SF, class domestic, grade 200, of PPP-B-636. Each fiberboard box shall be closed in accordance with method II as specified in the appendix of PPP-B-636. When unit pack quantities other than those specified in table V are required, buckles of the same class and size only shall be unit packed in containers conforming to PPP-B-566, PPP-B-676, or PPP-B-636. The containers shall be sealed or closed in accordance with the applicable container document.

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TABLE V. Unit packing and packing of buckles, end bar with tongue

Size (inches)	Quantity per unit pack	Quantity per pack
Class 1:		
5/8	800	4800
3/4	650	3900
7/8	500	3000
1	400	2400
1 1/4	300	1800
1 3/8	250	1500
1 1/2	240	1440
Class 3:		
1	300	1800
1 1/4	300	1800

5.1.2 Commercial. The buckles shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Unless otherwise specified (see 6.2), buckles of one class and size only, preserved as specified in 5.1, shall be packed in the quantities specified in table V in containers conforming to overseas type of PPP-B-601 or class 2, style 4 of PPP-B-621. Each container shall be closed and reinforced in accordance with the applicable container specification and waterproofed with a sealed case liner conforming to type I or II, grade C of MIL-L-10547.

5.2.2 Level B packing. Unless otherwise specified (see 6.2), buckles of one class and size only, preserved as specified in 5.1, shall be packed in the quantities specified in table V in a snug-fitting fiberboard shipping container conforming to style FTC, grade V3c or V3s of PPP-B-636. Each shipping container shall be closed and reinforced in accordance with the appendix of PPP-B-636.

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5.2.3 Commercial packing. Buckles, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), buckles packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type I or Ia, as applicable, of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking loads shall be effected by reversing the pattern of each course. If the container is of a size that does not conform to any of the patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The buckles are intended for use on leather items.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.

b. Class and size of buckle required (see 1.2 and Drawing 4-1-30).

c. When nickel plating is required (see 3.3.2).

d. Selection of the applicable level of preservation and packing (see 5.1 and 5.2).

e. When quantities other than those specified in table V are required (see 5.1.1, 5.2.1, and 5.2.2).

f. When palletized loads are required (see 5.4).

* 6.3 Recycled material. The contractor is encouraged to use recycled material when practical, provided that it meets the requirements of this document (see 3.1).

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6.4 Changes from previous issues. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue have been made. This was done as a convenience only, and the Government assumes no liability whatsoever for inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-3493D	Buckles, End Bar with Tongue (For Leather)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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